

Aztec Logic Module Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Aztec Labs on Aztec Logic Module. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated April 6, 2026, was delivered to Aztec Labs, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 4 days, the Cyfrin team conducted an audit on the Aztec Logic Module code provided by Aztec Labs. The findings consist of 4 Low severity issues with the remainder being informational.

Summary

Project Name	Aztec Logic Module
Repository	aztec-packages
Commit	e4712cda8def...
Audit Timeline	Mar 16th - Mar 19th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	4	2	2
Informational	4	1	3
Gas Optimizations	0	0	0
Total	8	3	5

6 Audit Scope

The audit scope was limited to the following files in the aztec-packages repository:

```
barretenberg/cpp/src/barretenberg/stdlib_circuit_builders/plookup_tables/uint.hpp
barretenberg/cpp/src/barretenberg/stdlib/primitives/logic/logic.cpp
barretenberg/cpp/src/barretenberg/stdlib/primitives/logic/logic.hpp
barretenberg/cpp/src/barretenberg/dsl/acir_format/logic_constraint.cpp
barretenberg/cpp/src/barretenberg/dsl/acir_format/logic_constraint.hpp
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The logic module implements circuit-friendly bitwise AND and XOR operations for the Aztec rollup. Noir's `&` and `^` operators on unsigned integers compile to `BlackBoxFunc::AND` / `XOR` ACIR opcodes, which Barretenberg processes via plookup-based constraints.

Production usage includes `is_power_of_2` checks (`n & (n-1) == 0`) in rollup structure validation and `compute_subtree_sizes` (`num_leaves & subtree_size`) in unbalanced Merkle trees.

8.1 Constraint Architecture

The module uses precomputed lookup tables (LogUp argument, called "plookup" in the codebase for historical reasons) with a two-level decomposition:

1. **32-bit chunks** (`logic.cpp`): the input is split into chunks of up to 32 bits. Each chunk is processed independently, and the results are reconstructed using positional scaling factors.
2. **6-bit slices** (plookup multi-table): each 32-bit chunk is further decomposed into [6, 6, 6, 6, 6, 2]-bit slices. Each slice is looked up in a BasicTable of 64 × 64 = 4096 entries (for 6-bit slices) or 4 × 4 = 16 entries (for 2-bit slices).

The plookup system has two table layers:

- **MultiTable** (metadata): describes how to decompose an input into slices. Created once at static initialization for all table types (UINT8/16/32/64 XOR/AND) and contains slice sizes, basic table IDs and function pointers.
- **BasicTable** (rows): the lookup entries (e.g., 64 × 64 = 4096 rows for a 6-bit table). Created lazily by the circuit builder only when a plookup read references that table.

8.2 LogUp Lookup Argument

LogUp (Log-Derivative Lookup) is the lookup argument used in Barretenberg's UltraHonk proving system. It replaces the original plookup grand-product approach with a sum-of-inverses identity. In the logic module, LogUp verifies that every 6-bit slice produced by the accumulator decomposition exists as a valid row in the XOR/AND BasicTable.

9 Attestation

Cyfrin attests that it performed a security review of the Aztec Logic Module code provided by Aztec Labs.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-04-06-cyfrin-aztec-logic-module-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Aztec Labs has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=1ef2a29433d5b89c4be7f2cce75ebcfd9357d031c31081de63df0d97f95f160e
F=2026-04-06-cyfrin-aztec-logic-module-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.